Monochromatic Rectangle Free Grids

A Design Theory Approach to the Erdős Box Problem

Jonathan Davidson

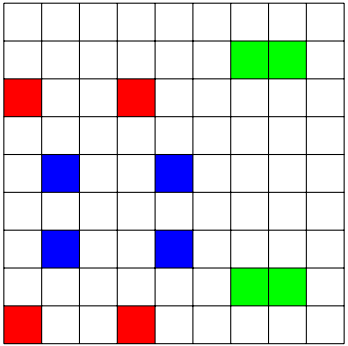
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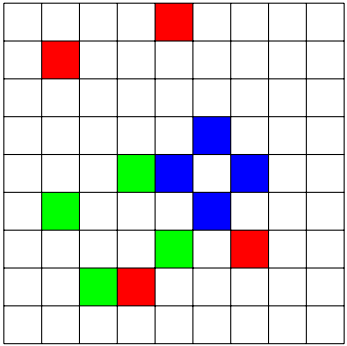
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Monochromatic Rectangles (MCR)



Non-examples



The Game

The rules

- You are given c colors.
- You only have a rows.
- How many columns can you color before you get an MCR?
- The longest grid wins!

The number of columns in the longest grid will be called the **extremal number**, denoted $\mathrm{ex}(a; c)$.

Extremal Combinatorics

How large does a structure become until we see a specific type of substructure?

The Role of Constructions

When solving extremal problems, it is important to figure out if a construction will be an upper bound or a lower bound on the extremal number.

- Given an MCR-free grid, there might be a longer one.
- Therefore, constructions will provide lower bounds.

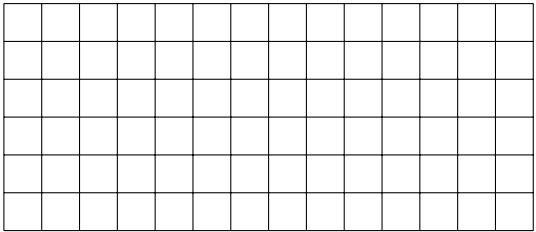
The Strategy

Steps to find extremal number:

1. Use theory to find upper bounds.
2. Use constructions to find lower bounds.
3. Improve until the bounds meet.

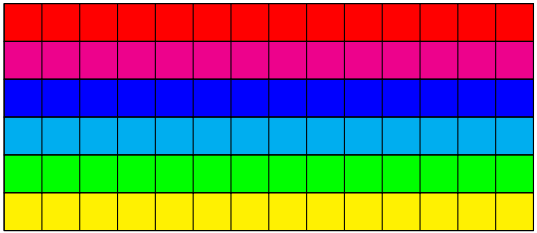
Rows $\leq$ Colors

You are given 6 colors and 6 rows. What is the longest MCR-free grid?



Rows $\leq$ Colors

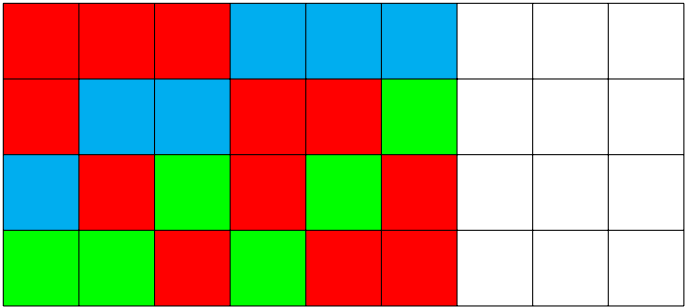
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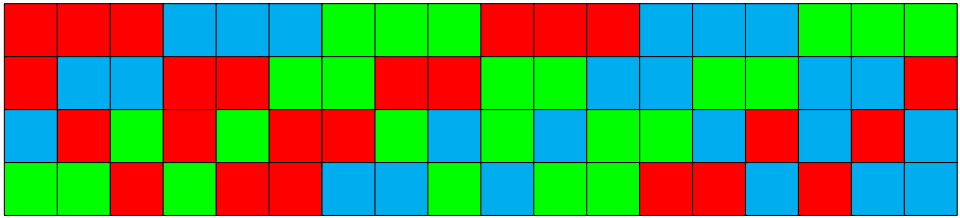
Rows = Colors + 1

You are given 3 colors and 4 rows. What is the longest MCR-free grid?

Rows = Colors + 1



Rows = Colors + 1



Key Observations

This example illustrates some key ideas:

- The pigeonhole principle guaranteed at least one pair in each column.
- If the same monochromatic pair occurs twice, we have an MCR.
- We can repeat the same monochromatic pair with different colors.
- Once we have seen all monochromatic pairs, we are optimal!

Upper Bound

Key Insight

If we encounter the same monochromatic pair twice, we will have an MCR. Lets count the total monochromatic pairs!

$$(\#\text{ MC pairs per col})(\#\text{ cols}) \leq (\#\text{ MC pairs}) = (\#\text{ colors})(\#\text{ pairs})$$

Upper Bound

Upper Bound

The extremal number of a grid with a rows and c colors satisfies

$$\mathrm{ex}(a;c) \leq \left\lfloor \frac{c}{\mathrm{mc}_1(a;c)} \binom{a}{2} \right\rfloor$$

where $\mathrm{mc}_1(a;c)$ denotes the minimum number of monochromatic pairs per column with a rows and c colors.

Two Questions

1. How do we find the minimum number of monochromatic pairs per row?
2. Can we find constructions that meet the upper bound?

Counting Pairs Pigeonhole Principle

Theorem (CP3)

Suppose there are a rows and c colors, where $a > c$. Then,

$$\mathrm{mc}_1(a; c) = r\binom{q+1}{2} + (c-r)\binom{q}{2} = c\binom{q}{2} + rq$$

where $a = qc + r$ and $0 \leq r < c$.

Counting Pairs Pigeonhole Principle

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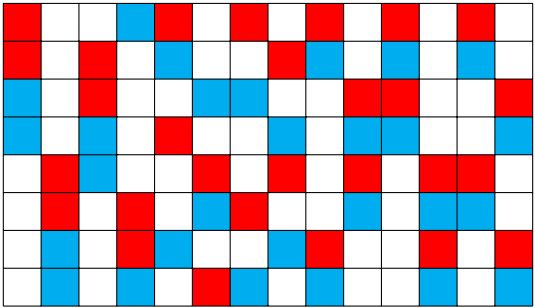
Rows = Colors + 2

You are given 6 colors and 8 rows. What is the longest MCR-free grid?

The upper bound yields

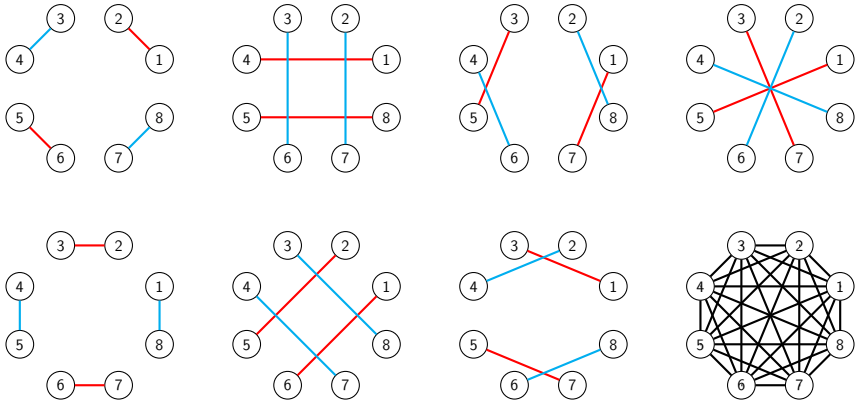
$$\mathrm{ex}(a; c) \leq \frac{6}{\mathrm{mc}_1(8,6)} \binom{8}{2} = \frac{6}{2} \cdot \frac{8(7)}{2} = 84$$

Rows = Colors + 2



Cycle through the 6 colors to produce a total of 84 columns.

2-Matching Decomposition of K_8



Rows = Colors + ℓ

Suppose there are $a = c + \ell$ rows and c colors where $1 \leq \ell < c$.

- There will be $\mathrm{mc}_1(c + \ell; c) = l$ monochromatic pairs per column.
- The row indices of each monochromatic pair correspond to an edges on the complete graph $K_{c+\ell}$.
- Each collection of row pairs in a column correspond to an l-matching.
- A maximal construction corresponds to an ℓ-matching decomposition of the complete graph $K_{c+\ell}$.

Existence of ℓ-Matching Decompositions

Theorem (Folkman-Fulkerson 1969)

If $a_1, a_2, \ldots, a_m$ is any sequence of integers such that $0 \leq a_i \leq a/2$ and $\sum_i a_i = \binom{a}{2}$, then the complete graph K_a can be decomposed into m matchings, where the ith matching contains a_i edges, $i = 1, \ldots, m$.

Existence of ℓ-Matching Decompositions

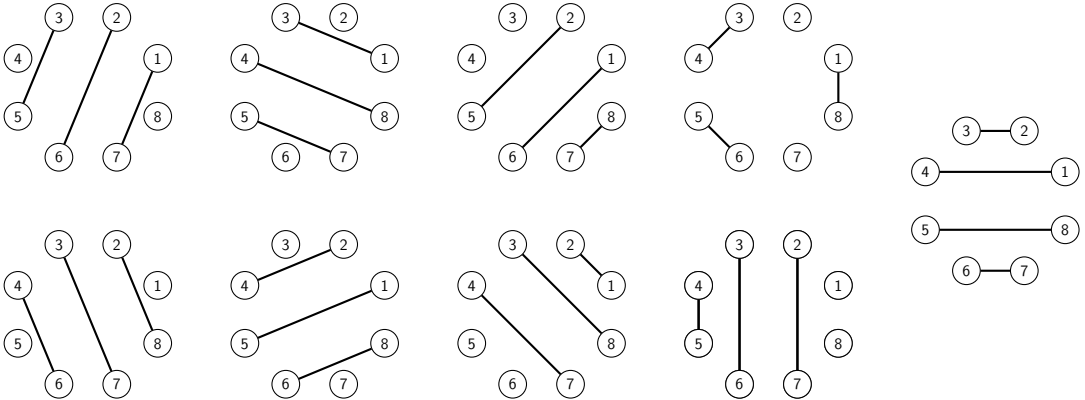
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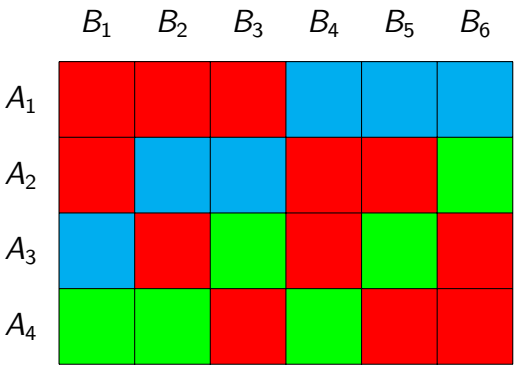
Proof Sketch.

1. Use Vizing's Theorems to get an edge coloring of K_a.
2. Transfer edges between matchings until the desirable sizes are achieved.

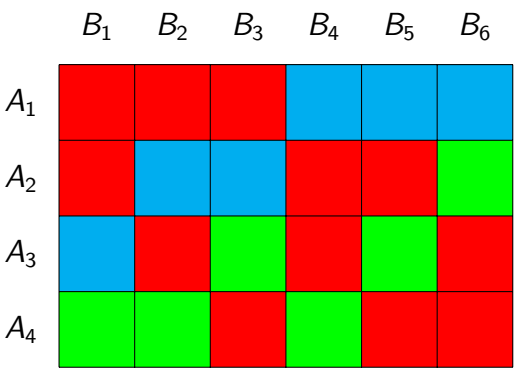
Near 3-Matching Decomposition of K_8

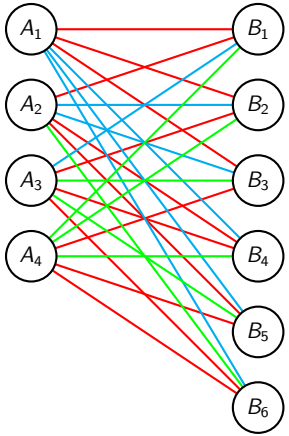


Grids and Bipartite Graphs

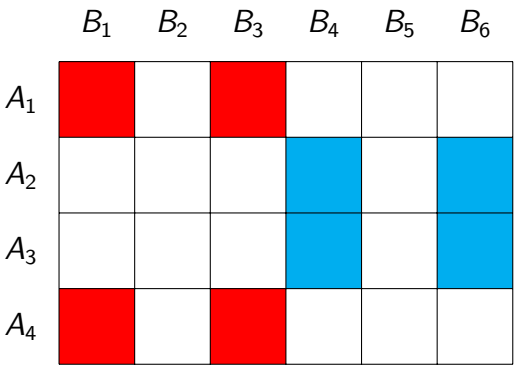


Grids and Bipartite Graphs

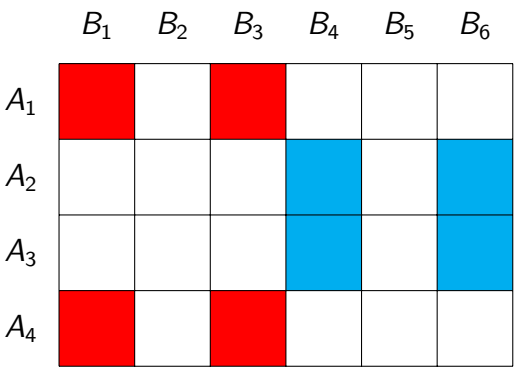


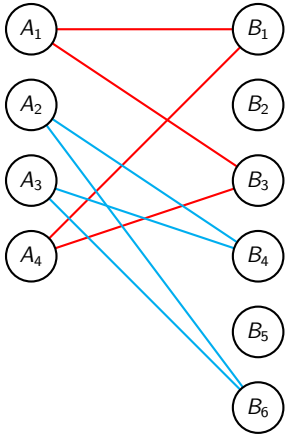


MCR and $K_{2,2}$



MCR and $K_{2,2}$





The Zarankiewicz Problem

The **Zarankiewicz number** $z(m, n; s, t)$ denotes the maximum edges on a bipartite graph $G = (A \cup B, E)$ such that

- $|A| = m$ and $|B| = N$
- G does not contain $K_{s,t}$

The Zarankiewicz Problem ask for a formula, or just asymptotic estimates, for $z(n, m; s, t)$.

The Zarankiewicz Problem

In the context of grid coloring:

- $z(m, n; 2, 2)$ is equivalent to finding the largest rectangle free subset of an $m \times n$ grid.
- MCR-free 2-colorings of grids yield exact values for Zarankiewicz numbers.

Estimates for Zarankiewicz Numbers

Theorem (Kővári-Sós-Turán)

For all $s \leq t$.

$$z(n, n; s, t) = O(n^{2-1/s})$$

There exists a construction $\Omega(n^{2-1/s})$ when $t > (s-1)!$ due to Alon, Kollár, Ronyai and Szabó.

Generalizing Zarankiewicz

We can generalize the Zarankiewicz problem in two ways:

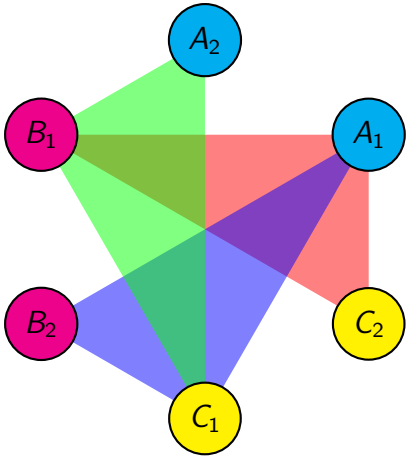
- Use more than 2 colors.
- Analyze hypergraphs.

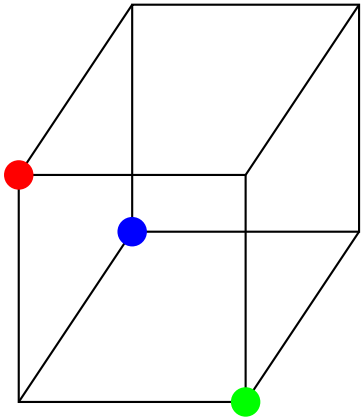
Using both generalizations leads to the multicolored Erdős Box Problem. This field of mathematics is called **Multipartite Hypergraph Ramsey Theory**.

3-Uniform Triartite Graphs

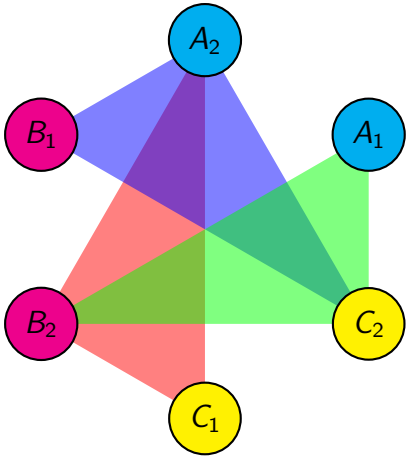
Each point in the $n_x \times n_y \times n_z$ grid represents a 3-hyperedge (x, y, z) in the 3-partite complete hypergraph, denoted $K^{(3)}_{n_x,n_y,n_z}$.

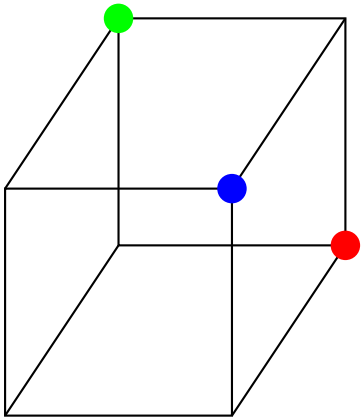
The 8 Hyperedges of $K^{(3)}_{2,2,2}$



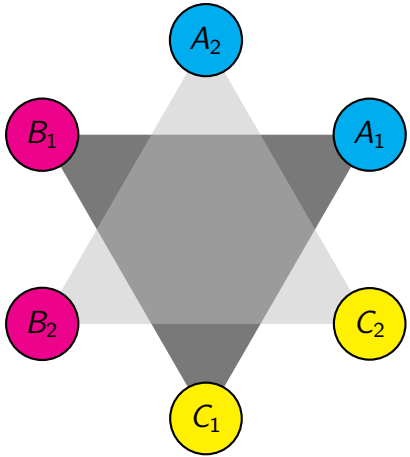


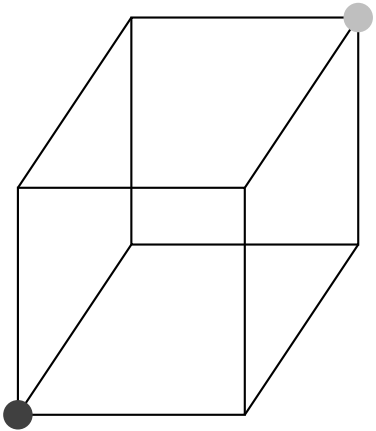
The 8 Hyperedges of $K^{(3)}_{2,2,2}$





The 8 Hyperedges of $K^{(3)}_{2,2,2}$





Multicolor Erdős Box Problem in 3D

Given a_1 rows, a_2 columns, and c colors, how many layers of these grids can we color before we are guaranteed a monochromatic box? This value is denoted $\mathrm{ex}_3(a_1, a_2; c)$.

Multicolor Erdős Box Problem in 3D

Given a_1 rows, a_2 columns, and c colors, how many layers of these grids can we color before we are guaranteed a monochromatic box? This value is denoted $\mathrm{ex}_3(a_1, a_2; c)$.

Equivalently, what is the largest value of a_3 such that all colorings of $K^{(3)}_{a_1,a_2,a_3}$ do not contain a monochromatic $K^{(3)}_{2,2,2}$.

Upper Bound 3D

Theorem

The extremal number of a 3D grid with length a_1, width a_2 and c colors satisfies

$$\mathrm{ex}_3(a_1, a_2; c) \leq \left\lfloor \frac{c}{\mathrm{mc}_2(a_1, a_2; c)} \binom{a_1}{2} \binom{a_2}{2} \right\rfloor$$

where $\mathrm{mc}_2(a_1, a_2; c)$ denotes the minimum number of monochromatic rectangles per layer with layer dimensions $a_1 \times a_2$ and c colors.

Estimates for EBP

Another direction to analyze grid colorings is by varying all directions at once.

Theorem (Erdős 1964)

$$\mathrm{ex}_d(n;2) = O\left(n^{d-\frac{1}{2^d-1}}\right)$$

This problem has been studied extensively via the random algebraic method.

Estimates for EBP

Theorem (Gunderson-Rödl-Sidorenko 1999)

Let s be the smallest positive integer such that $(sd-1)/(2^d-1)$ is an integer.

$$\mathrm{ex}_d(n;2) = \Omega\left(n^{d-\frac{d-1/s}{2^d-1}}\right)$$

Theorem (Conlon-Pohoata-Zakharov 2021)

Let r and s be positive integers such that $d(s-1) < (2^d-1)r$.

$$\mathrm{ex}_d(n;2) = \Omega\left(n^{d-\frac{r}{s}}\right)$$

t-designs

Definition

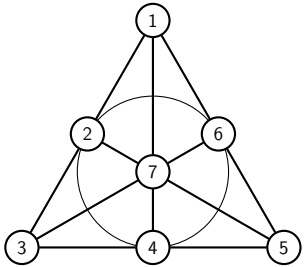
A t-design is a family of subsets called blocks with ground set v.

- A design is **balanced** if each t-subset occurs in λ blocks.
- A design is **uniform** all blocks have the same size k.
- A design is **resolvable** if the blocks can be partitioned into parallel classes that partition the ground set.

A uniform, balanced 2-design is called a balanced incomplete block design, denoted BIBD(v, k, λ).

The Fano Plane

The Fano plane is a uniform, balanced design with block size 3.



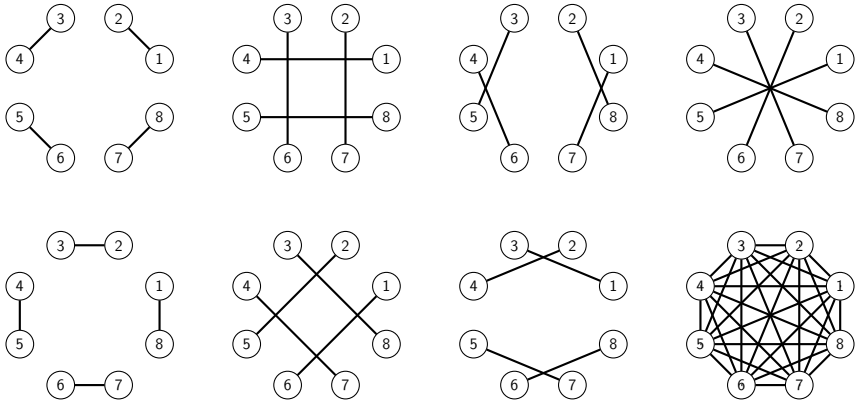
$\{1,2,3\}$ $\{3,4,5\}$ $\{5,6,1\}$

$\{1,7,4\}$ $\{2,7,5\}$ $\{3,7,6\}$

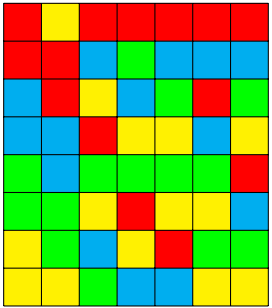
$\{2,4,6\}$

1-factorization of K_8

The 1-factorization of K_8 is a resolvable, uniform, balanced 2-design with block size 2.

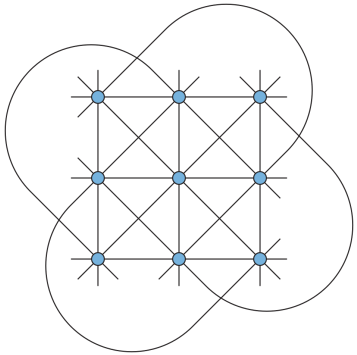


Rows = 2 * Colors



Hesse Configuration

The Hesse configuration is a resolvable, uniform, balanced 2-design with block size 3.



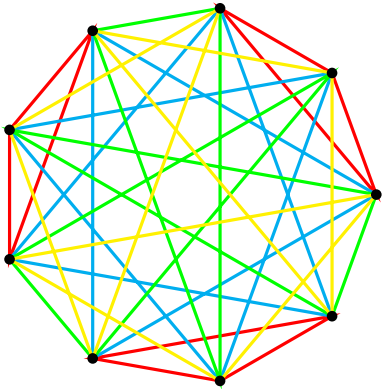
$\{1,2,3\}$ $\{4,5,6\}$ $\{7,8,9\}$

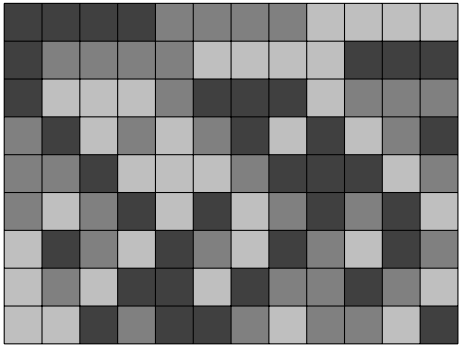
$\{1,4,7\}$ $\{2,5,8\}$ $\{3,6,9\}$

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$\{1,6,8\}$ $\{2,4,9\}$ $\{3,5,7\}$

Rows = 3 * Colors





Rows $= \ell$ * Colors

Theorem

If there exists a resolvable BIBD$(\ell c, \ell, 1)$ *for some integer* ℓ*, then*

$$\mathrm{ex}(\ell c; c) = c\frac{\binom{\ell c}{2}}{\binom{\ell}{2}}$$

Existence of Designs

Theorem (Ray-Chaudhuri-Wilson (1971))

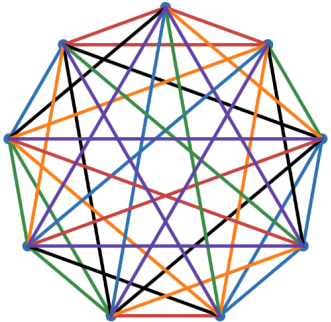
There exists a resolvable bibd$(v, 3, 1)$ *if and only if* $v \equiv 3 \mod 6$.

Theorem (Hanani-Ray-Chaudhuri-Wilson (1971))

There exists a resolvable bibd$(v, 4, 1)$ *if and only if* $v \equiv 4 \mod 12$.

Further Work

Class Uniformly Resolvable Pairwise Balanced Design, CURD($\{2, 3\}, 9, 3$).



Further Work

This can be thought of as a resolvable CURD($\{1, 2\}, 8, 3$) with $\lambda = 5$.

